6. Project Testing Phase

6.1 Introduction

Testing verifies that EduGenie satisfies the specified requirements and performs its functions correctly. It identifies errors, security issues, integration problems, and unexpected behavior. Each module is tested individually and then as an integrated system.

6.2 Objectives of Testing

- To verify that all requirements are implemented correctly.
- To identify and correct software errors.
- To ensure authentication works properly and protected data is inaccessible to unauthorized users.
- To verify that data is stored and retrieved correctly.
- To verify correct quiz scoring and dashboard statistics.
- To evaluate the quality and relevance of Gemini-generated content.
- To verify usability and reliability.

6.3 Testing Strategy

Unit Testing → Integration Testing → System Testing → User Acceptance Testing

6.4 Unit Testing

Individual functions are tested in isolation, including authentication functions, quiz scoring, prompt-building functions, response parsing, data validation, and database operations. For example, the quiz score is calculated as Score = (Correct Answers ÷ Total Questions) × 100, and different input combinations are tested to confirm the expected result.

6.5 Integration Testing

- Frontend and backend communication.
- Backend and database operations.
- Authentication and protected APIs.
- Backend and Gemini API.
- Quiz module and dashboard statistics.

6.6 System Testing

Complete workflows are tested: registration, login, asking the tutor a question, requesting an explanation, summarizing notes, generating and submitting a quiz, creating flashcards, generating a study plan, viewing history, viewing the dashboard, and logging out.

6.7 Functional Test Cases

Test ID	Function	Test Input / Action	Expected Result
FT01	Registration	Enter valid user details	Account is created
FT02	Registration	Enter incomplete details	Validation message appears
FT03	Login	Enter valid credentials	User is logged in

Test ID	Function	Test Input / Action	Expected Result
FT04	Login	Enter incorrect password	Login is rejected
FT05	AI Tutor	Ask a valid question	Relevant answer is displayed and saved
FT06	AI Tutor	Ask a follow-up question	Answer reflects earlier context
FT07	Explanation	Select topic and level	Level-appropriate explanation appears
FT08	Summarizer	Submit notes	Concise summary is generated
FT09	Summarizer	Submit empty text	Validation message appears
FT10	Quiz	Generate quiz on a topic	Requested number of questions is shown
FT11	Quiz	Submit answers	Correct score and explanations are shown
FT12	Flashcards	Generate flashcards	Term–definition cards are displayed
FT13	Study Plan	Enter subject, time, and date	Structured plan is generated
FT14	History	Open history page	Previous items are displayed
FT15	Dashboard	Open dashboard	Correct statistics and charts appear
FT16	Logout	Select logout	Session ends

6.8 Authentication and Security Testing

- Access to protected APIs without a token is rejected.
- Expired or invalid tokens are rejected.
- A user cannot view another user's history.
- Passwords are stored as hashes.
- The Gemini API key is not exposed to the browser.

6.9 AI Response Testing

Gemini responses are evaluated for relevance, clarity, correctness, and format. Sample questions from several subjects are used. Cases tested include empty or very long input, unclear questions, and malformed quiz output. Because AI-generated content can occasionally be inaccurate, the application advises learners to verify important information.

6.10 Performance and Usability Testing

Page load time, API response time, and behavior during Gemini delays are observed. Usability is checked for navigation, clarity of messages, and responsiveness on desktop and mobile screens.

6.11 User Acceptance Testing

A small group of students uses the application and provides feedback on ease of use, usefulness of explanations, and quality of quizzes and study plans. Reported issues are corrected before the final demonstration.

6.12 Testing Summary

After testing and correction of defects, all major modules were found to operate according to the requirements, and the application was considered ready for demonstration.